

Impact Statement Card

Intended Use

Improve document retrieval for a Retrieval-Augmented Generation system that supports smallholder farmers and agricultural extension workers seeking advice on crop production, pest/disease management, soil fertility, fertilizer application, irrigation, and climate adaptation.

Values Guiding the Design: Usefulness, relevance, transparency, responsibility, accountability.

Evidence of Impact

- TF-IDF baseline: $nDCG@5 = 0.5131$
- Dense retrieval: $nDCG@5 = 0.7856$

The gap between these shows why retrieval quality matters directly: better ranking means more relevant documents surface in the top 5, which is the evidence base a downstream RAG system would generate its answer from.

Limitations on Real-World Deployment

- The 695-document, 308-query dataset cannot establish that the system performs equally well across all agricultural communities, regions, or crops.
- Evaluation covers retrieval/ranking only — not the quality, safety, or accuracy of any final generated advice.
- Uneven relevance-label distribution (mostly 0s) means $nDCG@5$ should be read alongside qualitative error analysis, not as a complete picture.
- Before real-world use, further evaluation with agricultural experts and intended users (farmers, extension workers) is necessary.

Accountability

The team commits to remaining accountable to the people this system is meant to support by explicitly naming these limitations rather than presenting benchmark scores as proof of real-world readiness.